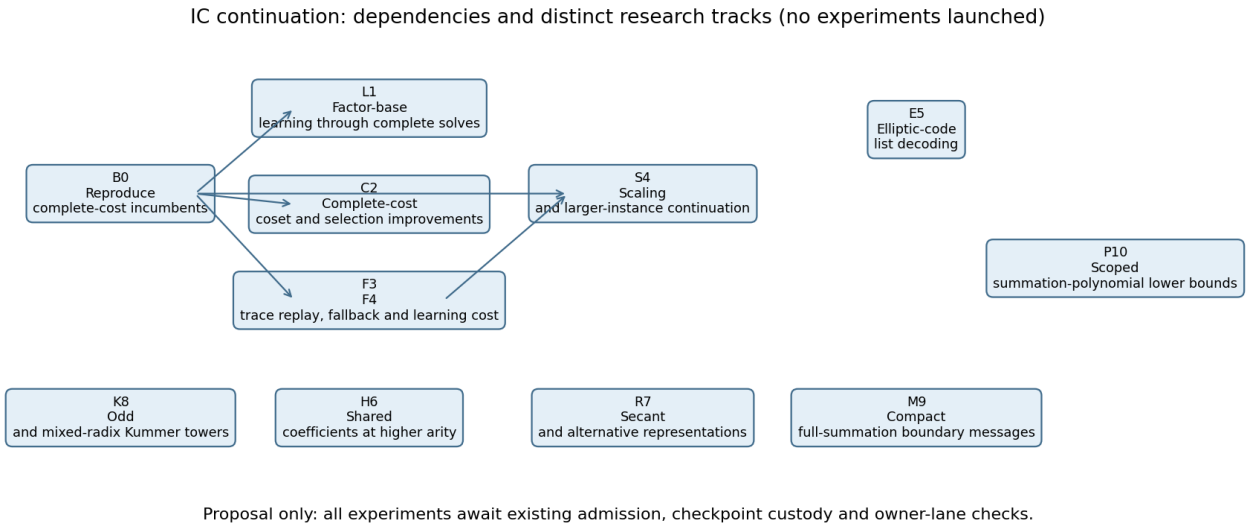
Index-calculus continuation program

5 November 2026 — preparation bundle, not scientific admission

All requested mechanisms have a concrete source-bound next step. The bundle does not create experiment approval, dispatch jobs, change official research status, or establish an impossibility theorem. Implementation and written derivations are producer assertions [pcc:unverified].



Proposed order: Reconcile B0; then at most two critical slots for F3 complete trace costs and L1 held-out learned bases. Resume K8 only in its owned lane and original protocol. E5 source review and P10 proof preparation proceed without experimental launches.

Admission: The WIP gate reports 91/12 open PRs and refuses expansion. Checkpoint upload access is absent. A scoped WIP exception was requested; no exception is assumed. No new cloud expenditure or resource expansion is authorized here.

Measurement: Cold one-target total includes construction, selection/training, failed queries, root discovery, signed/subgroup validation, independent rank, linear algebra, descent and verification. Reused-setup batches are separate. Counted work and isolated wall time are separate metrics.

B0 — Reproduce complete-cost incumbents

Question: Do the named baseline, strongest applicable solver and matched rho have complete and comparable accounting?

Next step: Bind existing study inputs, code and raw exports; independently replay one original complete panel before extending it. Preserve separate prime, ordinary-binary and Koblitz strata.

Controls: fresh-process A/A; known-answer logs and deliberate bad witness; strongest matched solver and rho on identical points

Scale: Original sizes first; no larger ladder before the reference discrepancy is resolved.

Deliverables: baseline-lock.json with curve/subgroup/base UIDs, code/binary/protocol hashes and unit definitions; canonical exports with all historical exclusions; discrepancy report and independent review

If favorable: Freeze replayed incumbent per stratum; admit one same-instance challenger.

If null: Repair the specific reproducibility/accounting mismatch without rewriting history.

L1 — Factor-base learning through complete solves

Question: Does a frozen selector lower full-DLP cost on unseen curves and dimensions?

Next step: Reproduce the l=4 differing-pick result, restore the pilot-dropped n=23 scope under a new protocol, then evaluate a frozen rank-aware selector against the hand rule and best available solver.

Controls: hand rule; shuffled-label model; planted useful base feature through same learner; tied-pick A/A; no target-log or post-solve feature leakage

Scale: Reproduction n=19,23 l=4; next n=31,41 l=5,6 subject to measured preflight. n>=43 only after complete lower rungs.

Deliverables: grouped train/validation/test split by curve and base family; selection search/training cost ledger; complete cold solves and held-out selector decisions

If favorable: Replicate on a second host/seed panel; ratchet the complete-cost baseline in the same stratum.

If null: Retain model diagnostics; test whether rank or descent dominates rather than resampling favorable cells.

C2 — Complete-cost coset and selection improvements

Question: Does staged selection retain its complete-cost improvement as construction and subgroup costs grow?

Next step: Replay P8d with its exact accounting, then prospectively compare legacy, staged selection, scan-all and k=1 versus k=16 jumps.

Controls: selection bypass A/A; scan-all expensive positive workload; legacy; fixed-base comparison isolating selector cost

Scale: Original band-17 family then at least three growing field-size bands; h=1 primary and h>1 separate.

Deliverables: same-instance selector/solve export; construction rejection and no-usable-base rows; per-phase attribution

If favorable: Promote only the matched complete-cost gain; feed frozen construction into the scaling arm.

If null: Keep the lower-scale gain; locate the cost crossover and retain the original domain coverage.

F3 — F4 trace replay, fallback and learning cost

Question: Is trace reuse cheaper including learning, failed replay, recovery and actual witness extraction?

Next step: Resume the original trace comparison before transfer. Run fresh and replay with identical systems; log each attempted replay and mandatory fresh fallback.

Controls: A/A fresh solves; same-system replay positive control; wrong-field/base/arity trace rejected; changed-target stress; SAT and UNSAT strata with natural mixture

Scale: Original prime-field toy rows separately from binary n17,19 l3,4; extend n23,31 and m3,4 only after exactness. Report one-target cold and separate batches T=8,32.

Deliverables: trace-source and invalidation identities; trace learning/cache/fallback ledger; separate counted-work and isolated-wall comparisons

If favorable: Freeze trace policy and price it in a complete solve; then transfer once to a new family.

If null: Keep exact rejection/fallback behavior; determine whether fill-in or trace instability explains the loss.

S4 — Scaling and larger-instance continuation

Question: Which phase and resource grows, and does a measured improvement persist out of sample?

Next step: Reproduce k16..24 without changing its frozen interpretation; freeze a prospective k26,28,30 continuation only after per-arm memory/runtime pilots and account budget readback.

Controls: matched rho; enumeration/grid on tiny cells; same-base dense representation; A/A builds; planted structure distinct from natural target distribution

Scale: Prime subgroup bits and binary field degree are separate axes. Proposed k26,28,30 and binary n23,31,41. No >128GiB cloud submission under current namespace policy; larger closure cells require an explicitly authorized capacity change.

Deliverables: phase-by-phase counts and matrix width/nnz; complete-panel fits with uncertainty and held-out-rung prediction; resource frontier retaining censored cells

If favorable: Add one next rung prospectively and independently replay; fit scaling only within the same algorithm and units.

If null: Publish the measured phase bottleneck and target a specific algorithmic change; never infer asymptotics from a flat finite segment.

E5 — Elliptic-code list decoding

Question: Does the actual received-word construction yield relations more cheaply than enumeration?

Next step: Retrieve and read the full Zhang-Liu ePrint/ProvSec construction; write exact divisor, code, received-word, radius, guessing and rationality assumptions before implementing it.

Controls: known subset-sum codeword; zero-word radius failure negative control; plain subset enumeration; random matched code/target and same code parameters

Scale: Exact tiny p first; planned p~2^12,2^14,2^16 with matched B and k=2..6 only after correctness and measured preflight.

Deliverables: primary-source theorem/assumption map; exact relation-codeword correspondence; tiny exhaustive checker and prospective cost protocol

If favorable: Freeze the decoder and compare per independent relation on a growing ladder before a full-DLP experiment.

If null: State the exact received-word/parameter obstruction; leave other decoding algorithms and parameter regimes open.

H6 — Shared coefficients at higher arity

Question: Can shared coefficients reduce complete solving work without hiding degree or rationality constraints?

Next step: Prove exact signed-model-set equivalence for compact selectors at m=3,4; retain m=2 only as a correctness/empty-chart negative control. Continue the source-bound F_p trail checker rather than an unadmitted one-hot expansion.

Controls: same-base per-root membership; dense membership; permutation quotient; empty squarefree m2 chart; repeated-root arm separate

Scale: Existing development fixtures, then s6..14 natural m3 panel as its own frozen protocol. No s15..27 continuation inherited automatically.

Deliverables: chart-complete signed tuple bijection; native root/witness extraction; source-bound solver explanations and UNSAT replay; setup/query/storage ledger

If favorable: Require prospective natural-query and reusable-setup slope gates from the owning protocol, then whole-DLP cost.

If null: Identify whether roots, selectors, explanation size or coefficient construction consumes the saving.

R7 — Secant and alternative representations

Question: Does lower handled degree reduce complete same-base work per independent relation?

Next step: Implement operational triangular/FGLM-style root discovery and share elimination across all four sign branches; do not use saved answers as a root oracle.

Controls: same-base dense system; matched nongraph secant; equal-size low-x/hash-x bases; tiny exhaustive root sets; repeated roots and poles

Scale: F17 exact controls, then prospective F83/F101/F127 d3..5 panel; higher m3,4 only after complete extraction and cost.

Deliverables: exact extractor with independent group replay; all chart/sign costs; unchanged historical caps plus new prospective results

If favorable: Advance to higher arity on independently checked complete savings across at least three growing cells.

If null: Preserve the degree theorem at its scope; reject only the tested performance construction.

K8 — Odd and mixed-radix Kummer towers

Question: Does a different radix reduce arithmetic/fill-in at matched membership information?

Next step: First resume frozen EXP-PKM-RADIX-001 with G0; do not alter its rows. Separately implement exponents 0,1,2 and cube rewrite rules for radix3 with exact solution-set tests before a new study.

Controls: radix2 incumbent; radix4 same-base encoding; curve-free shape null; small independent enumeration; associativity and reduction confluence on tiny quotient

Scale: Resume existing L2/L3/X2/A1 exactly. New odd-radix study begins at tiny sizes then >=3 increasing log2(base-size) rungs; fix field/curve where possible and cross the remaining mismatch.

Deliverables: G0 original-count regression; original weighted-grading export; radix3/mixed normal form and multiplication checker; same-base radix2/4 and matched-size 2/3 controls

If favorable: Use counted arithmetic and fill-in to choose a representation; then compare complete relation and full-DLP cost.

If null: Separate weighted-degree renaming from actual arithmetic; retain odd-radix construction if correct even when it loses.

M9 — Compact full-summation boundary messages

Question: Can an exact message reuse full constrained summation rather than local membership alone?

Next step: Define an explicit exact boundary representation over the rational two-isogeny fibre reference; implement specialization and prefix witness extraction with every addition chart.

Controls: direct full-sum reference; membership-only cache; full target/tuple table explicitly priced; identity/inverse/doubling/pole/kernel/subgroup cases

Scale: Exact tiny p exhaustive first; increase base size, m=2,3,4 and target count separately. T=1 cold primary; amortized setup at T=8,32 a separate comparison.

Deliverables: coefficient algebra and exact precision policy; signed ordered-tuple semantics including repetitions; message construction/storage profile; full small-field count and witness equivalence

If favorable: Price all intermediate states and roots before comparing against the strongest direct oracle.

If null: Identify the precise state/message explosion; do not infer a universal tensor-network lower bound.

P10 — Scoped summation-polynomial lower bounds

Question: Which quantified algorithm class can actually be excluded, and what proof bridges are missing?

Next step: Review the written counting lemmas in PROOF_PROGRAM.md, formalize feasible statements in the owning Lean library, and audit every quantified hypothesis before attempting a solver-family lower bound.

Controls: anomalous/weak curves excluded or labelled; auxiliary-variable low-degree membership counterexample; coordinate-oracle/query-bound counterexample; small finite exact counts; alternate solver allowed as scope test

Scale: General finite-group counting lemma first; then one exact solver/presentation family. No claim about all curves, all encodings or future algorithms.

Deliverables: proof_search_map with definitions and quantifiers; readable proofs plus Lean theorem/axiom map; counterexample and proves-too-much controls; explicit unresolved average-case and representation bridges

If favorable: Publish only a faithful scoped lower bound after independent proof/Lean review; state exactly which methods remain outside it.

If null: Preserve the failed conjecture and counterexample; refine the model without weakening the original claim silently.

What a no-go theorem would require

Define the theorem: Specify curve/subgroup family, target distribution, algorithm and presentation class, success probability, preprocessing/advice, memory, cost model, uniformity and asymptotic quantifiers. “Never finds a relation” is false; “never beats rho in this model” is a possible scoped target.

First written counting result: For a finite group H and ordered m-tuples from a base F, sum_R N_m(R)=|F|^m. Hence uniform-target success probability is at most min(1,|F|^m/|H|). This does not prove uniform coverage or independent row rank.

Restricted collection floor: With iid uniform queries, at least K successes needed, setup A, and a per-query lower bound c0, expected work is at least A+c0*K*|H|/min(|H|,|F|^m). The query/rank/cost assumptions must be proved for the actual algorithm. Adaptive policies and amortized tables require separate arguments.

Membership lemma: A nonzero univariate polynomial vanishing on B distinct field elements has degree at least B. Auxiliary-variable systems and towers remain outside this statement; PKM2016 explicitly uses that escape.

Substantial missing bridge: A solver-family theorem must establish an unavoidable charged-resource lower bound for every allowed parameter choice, then justify its probability over target instances. Degree lower bounds do not automatically imply runtime lower bounds; a finite fitted slope is not an asymptotic proof.

Countercontrols: Keep coordinate-aware algorithms outside generic-group conclusions; audit pulled-back degrees for large multipliers; do not infer k-SUM hardness from random-looking statistics. Test weak curves, alternate solvers and extended membership formulations against every proposed theorem.

Formalization and review: The readable statements, exact hypotheses and countermodels are in PROOF_PROGRAM.md. Finite Python development checks are supplied. Lean formalization, axiom audit, the algorithm bridge and independent proof review remain uncompleted; no universal theorem is asserted.

Pinned source index

radix-run: isogeny-labs/crypto @ f384852fdc2a / research/pkm_radix_grading_20261104/run_cells.sh

Open immutable source

radix-protocol: isogeny-labs/crypto @ f384852fdc2a / research/pkm_radix_grading_20261104/protocol.json

Open immutable source

atlas-cost: isogeny-labs/crypto @ f384852fdc2a / research/curve25519_ic_atlas_20261104/REPORT.md

Open immutable source

trace-run: isogeny-labs/ml-cryptanalysis @ eb227d3ccd93 / experiments/f4-trace-replay/run_cell.py

Open immutable source

trace-protocol: isogeny-labs/ml-cryptanalysis @ eb227d3ccd93 / experiments/f4-trace-replay/PROTOCOL.md

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learning-null: isogeny-labs/ml-cryptanalysis @ eb227d3ccd93 / experiments/fb-cost-model-e2e-l4/report.md

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complete-cost: isogeny-labs/ml-cryptanalysis @ eb227d3ccd93 / experiments/prime-coset-domain-20261031/report_p8d.md

Open immutable source

solver-reference: isogeny-labs/ml-cryptanalysis @ eb227d3ccd93 / experiments/published-ic-reference-arm/README.md

Open immutable source

boundary: isogeny-labs/crypto-autoresearcher @ 5440a26b3669 / knowledge/open-problems/KN-OPEN-08ba7e.md

Open immutable source

decoding-desk: isogeny-labs/crypto-autoresearcher @ 5440a26b3669 / docs/reports/ecdlp-leads-desk-20261104/README.md

Open immutable source

shared-lead: isogeny-labs/crypto-autoresearcher @ 5440a26b3669 / knowledge/leads/ecdlp/ic-prime/LEAD-ICP-shared-coefficient-membership.yaml

Open immutable source

secant-lead: isogeny-labs/crypto-autoresearcher @ 5440a26b3669 / knowledge/leads/ecdlp/ic-prime/LEAD-ICP-split-graph-secant-presentation.yaml

Open immutable source

binary-degree: isogeny-labs/crypto-autoresearcher @ 5440a26b3669 / knowledge/leads/ecdlp/ic-engine/LEAD-ICE-descent-closure-degree-at-scale.yaml

Open immutable source

Primary literature: PKM2016 primary construction and experimental conclusions were read. Zhang-Liu ePrint 2018/795 PDF request returned HTTP 403; the actual received-word construction remains unread.

Graphs: This is a planning diagram. No measured research graph or leaderboard is modified, because there is no new independently verified scientific result.